

Milestone 1 – Web Application Development and Setup

1. Introduction

This project focuses on the development of a web application using the Flask framework. Milestone 1 concentrates on setting up the backend structure, defining routes, creating HTML templates, and validating application logic using local data storage. The objective of this milestone is to ensure the application functions correctly in a local environment before integrating cloud services in later milestones.

2. Objectives of Milestone 1

The main objectives of Milestone 1 are:

- To initialize a Flask backend application
 - To design basic web pages using HTML templates
 - To implement routing and request handling in Flask
 - To store and retrieve user data using Python lists and dictionaries
 - To test the application locally on a development server
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3. Technology Stack

The technologies used in this milestone include:

- **Programming Language:** Python
- **Backend Framework:** Flask
- **Frontend:** HTML
- **Data Storage:** Local Python lists and dictionaries
- **Deployment Environment:** Localhost (Flask development server)

4. System Architecture

The system follows a simple client–server architecture:

- The user interacts with the application through a web browser.
- HTTP requests are sent to the Flask backend.
- Flask processes the requests, performs operations on local data structures, and renders HTML templates.
- The response is returned to the browser for display.

This architecture helps validate application logic without relying on external databases or cloud services.

5. Application Structure

The project is organized into the following structure:

milestone1_app/

├─ app.py

├─ requirements.txt

├─ templates/

│ └─ index.html

│ └─ register.html

│ └─ login.html

└─ dashboard.html

- **app.py:** Contains the Flask application logic and route definitions
- **templates:** Holds HTML files rendered by Flask
- **requirements.txt:** Lists required Python packages

6. Implementation Details

6.1 Flask Application Setup

The Flask application is initialized in `app.py`. A secret key is configured to manage user sessions. The application runs using Flask’s built-in development server.

6.2 Routing and Templates

The following routes were implemented:

- / – Home page
- /register – User registration page
- /login – User login page
- /dashboard – User dashboard after successful login
- /logout – Ends the user session

Each route renders a corresponding HTML template using Flask's `render_template()` function.

6.3 Local Data Storage

User data is stored temporarily using a Python list of dictionaries. This approach allows testing of application logic such as registration and authentication without using a database.

6.4 Authentication and Sessions

Basic authentication logic is implemented by validating user credentials stored in the local list. Flask sessions are used to maintain the logged-in state of users and restrict access to the dashboard page.

7. Output and Results

The application was tested locally and produced the following results:

- The Flask server runs successfully on localhost
- Users can register and log in using the web interface
- Authenticated users are redirected to a dashboard
- Session management ensures secure navigation between pages

Screenshots of the application output, including the home page, registration page, login page, and dashboard, are included as part of the project deliverables.

8. Limitations

- Data is stored only in memory and is lost when the server restarts
 - No database or cloud services are integrated in this milestone
 - Security features are basic and intended only for development purposes
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9. Conclusion

Milestone 1 successfully establishes the foundation of the web application by implementing a Flask backend with routing, templates, and local data handling. The application runs smoothly in a local environment and meets all milestone requirements. This implementation provides a strong base for future milestones involving cloud integration and advanced features.

10. Future Scope

In subsequent milestones, the application will be enhanced by:

- Integrating AWS services such as DynamoDB
 - Implementing secure authentication mechanisms
 - Deploying the application to a cloud environment
 - Enhancing scalability and performance
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Milestone 1 Status: Completed